

BANARAS HINDU UNIVERSITY

B.A./B.Sc. (Hons.) Semester VI Examination, 2013-14

Paper No. STA/STB-603: Elements of Stochastic Processes

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer any **five** questions. All questions carry equal marks. Symbols have their usual meanings.*

Question 1.

What do you mean by a stochastic process? Discuss its classification according to state space and time domain. Give one example of each type. [14]

Question 2.

Describe the Gambler's Ruin Problem. Obtain the expression for the expected duration of the game for $p = \frac{1}{2}$ and for $p \neq \frac{1}{2}$. [14]

Question 3.

- (a) Define a Markov chain. Give two examples. [7]
- (b) Define the n -step transition probability. Also prove the Chapman–Kolmogorov equation:

$$P_{ij}^{(n+m)} = \sum_k P_{ik}^{(n)} P_{kj}^{(m)}.$$

[7]

Question 4.

- (a) Define persistent, transient, ergodic, and essential states of a Markov chain. [7]
- (b) Describe stationary distribution for a Markov chain. Give conditions under which it exists. [7]

Question 5.

- (a) Describe the time-dependent Poisson process. Obtain the expression for $P_n(t)$, mean and variance. [7]
- (b) If $\{X(t), t \geq 0\}$ is a Poisson process, obtain $P[X(s) = k \mid X(t) = n]$ for $s < t$. [7]

Question 6.

Describe the pure death process. Obtain the expression for $P_n(t)$ in the case of a linear death process. [14]

Question 7.

Describe the linear birth–death process. Derive the expression for its probability generating function. [14]

Question 8.

Describe the Galton–Watson branching process. For this process, prove that the probability generating function satisfies $P_n(s) = P_{n-1}(P(s)) = P(P_{n-1}(s))$, where $P_n(s) = \sum_{k=0}^{\infty} P(Z_n = k) s^k$. [14]